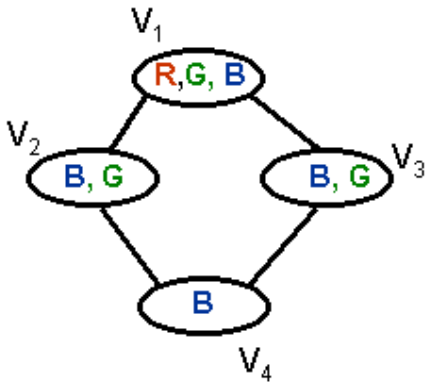
Answers to Tutorial #9: Constraint Satisfaction Problem

Simple Map Colouring Problem

Let's look at a simple map coloring problem and explore the behavior of constraint propagation, backtracking and forward checking in this context. Here is the relevant problem specification:



1. The type of graph shown above is known as a constraint graph for the CSP.
2. Write the values in each of the indicated variable domains **after** any changes required to achieve arc consistency for just that arc. Then, assume that the following arcs are done sequentially, with the effects on the domains propagating. Enter domains as a sequence of letters, for example, `R B`, do not include parenthesis, do not spell out colour names, leave spaces between the letters. If there are no values left in a domain, enter `none`.

V_1 - V_2: D_1=R,G,B D_2= B,G
V_1 - V_3: D_1= R,G,B D_3= B,G
V_2 - V_4: D_2= G D_4= B
V_3 - V_4: D_3= G D_4= B
V_1 - V_2: D_1= R, B D_2= G
V_1 - V_3: D_1= R, B D_3= G

3. Assume we use pure backtracking to search for a solution to this problem. We use a fixed variable ordering in the search (V_1, V_2, V_3 and V_4) and the values are considered in the order shown in the picture above. Then, show the order in which assignments are considered by backtracking (this is analogous to the order in which nodes are expanded by depth-first search). Show the assignment even if it is immediately found to be inconsistent upon testing. Assume that the search continues even after finding a valid solution. Enter each assignment (one assignment in each of the spaces below) as a number from 1 to 4 (indicating the variable) followed by a letter, drawn from `R, G, B`, leave a space between them, e.g. `1 R`:
   a. 1R
   b. 2B
   c. 3B
   d. 4B
   e. 3G
   f. 4B
   g. 2G
   h. 3B
   i. 4B

j. 3G

4. Repeat assuming we use backtracking **with forward checking** to search for a solution to this problem. We use the same variable ordering and value ordering as before. Show the order in which assignments are considered by BT-FC. Whenever propagating after an assignment causes a domain to become empty, that causes backup in the search. Assume that the search continues even after finding a valid solution. Enter each assignment (one assignment in each of the spaces below) as before:
   a. 1R
   b. 2B
   c. 2G
   d. 3B
   e. 3G
   f. 4B
   g. 1G
   h. 2B
   i. 1B
   j. 2G